# Advanced Power System Analysis

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# 1

# 2 Introduction

In this document, I review two papers. The first papers introduces the derivation is for three-phase, balanced systems. The second derviation is for three-phase unbalanced systems.

## 2.1 First Paper: Network Reconfiguration in Distribution Systems for Loss Reduction and Demand Balancing, Baran et. al.

## 2.2 Paper's Objectives

1- How can we change the network configuration to reduce resistive losses? 2- How can the load be distributed such that no overloads occur on the T-Lines?

# 3 Background Information

**NOTE: This file contains supporting matrix algebra and the fuller multiphase branch-flow formulation of Gan and Low (Gan and Low 2014). The working equations are in `ch2_voltage_reactive_equations.qmd`.**

## 3.1 Three-phase Branch Power Matrix

Recall that $\bar{\mathbf{V}}_i$ and $\bar{\mathbf{I}}_{ij}$ are $3 \times 1$ column vectors. We previously used the per-phase complex power vector:

$$\bar{\mathbf{s}}_{ij} = \begin{bmatrix} P_{ij}^a + jQ_{ij}^a \\ P_{ij}^b + jQ_{ij}^b \\ P_{ij}^c + jQ_{ij}^c \end{bmatrix}$$

Gan and Low also use the full branch power matrix:

$$\bar{\mathbf{S}}_{ij} = \bar{\mathbf{V}}_i \bar{\mathbf{I}}_{ij}^H \tag{3.1}$$

Expanding the matrix multiplication:

$$\bar{\mathbf{S}}_{ij} = \begin{bmatrix} \bar{V}_i^a (\bar{I}_{ij}^a)^* & \bar{V}_i^a (\bar{I}_{ij}^b)^* & \bar{V}_i^a (\bar{I}_{ij}^c)^* \\ \bar{V}_i^b (\bar{I}_{ij}^a)^* & \bar{V}_i^b (\bar{I}_{ij}^b)^* & \bar{V}_i^b (\bar{I}_{ij}^c)^* \\ \bar{V}_i^c (\bar{I}_{ij}^a)^* & \bar{V}_i^c (\bar{I}_{ij}^b)^* & \bar{V}_i^c (\bar{I}_{ij}^c)^* \end{bmatrix}$$

Note that the **diagonal elements** are the conventional complex powers for each phase:

$$\text{diag}(\bar{\mathbf{S}}_{ij}) = \bar{\mathbf{s}}_{ij}$$

The **off-diagonal elements** are cross-phase voltage-current products. They do not represent extra downstream branches or separate real/reactive phase powers. The full $3 \times 3$ matrix is useful for Gan and Low's branch-flow and optimization formulations, but it is not required just to substitute phase power into KVL.

## 3.2 Derive the Magnitude of the Voltage

Following the same approach as in Equation 4.20, we can obtain squared voltage magnitudes using the three-phase KVL equation:

$$\bar{\mathbf{V}}_j = \bar{\mathbf{V}}_i - \bar{\mathbf{Z}}_{ij}\bar{\mathbf{I}}_{ij}$$

To retain all cross-phase voltage products, multiply the equation on the right by its **Hermitian transpose**:

$$\bar{\mathbf{V}}_j\bar{\mathbf{V}}_j^H = (\bar{\mathbf{V}}_i - \bar{\mathbf{Z}}_{ij}\bar{\mathbf{I}}_{ij})(\bar{\mathbf{V}}_i - \bar{\mathbf{Z}}_{ij}\bar{\mathbf{I}}_{ij})^H \tag{3.2}$$

The Hermitian transpose, denoted by $H$, means **conjugate transpose**; it does **not** mean extracting the diagonal. A $3 \times 1$ voltage vector times its $1 \times 3$ conjugate transpose produces a $3 \times 3$ matrix.

### 3.2.1 Solving for the left hand side of Equation 3.2

$$\bar{\mathbf{V}}_j\bar{\mathbf{V}}_j^H = \begin{bmatrix} \bar{V}_j^a \\ \bar{V}_j^b \\ \bar{V}_j^c \end{bmatrix}_{3\times1} \begin{bmatrix} (\bar{V}_j^a)^* & (\bar{V}_j^b)^* & (\bar{V}_j^c)^* \end{bmatrix}_{1\times3} = \begin{bmatrix} |\bar{V}_j^a|^2 & \bar{V}_j^a(\bar{V}_j^b)^* & \bar{V}_j^a(\bar{V}_j^c)^* \\ \bar{V}_j^b(\bar{V}_j^a)^* & |\bar{V}_j^b|^2 & \bar{V}_j^b(\bar{V}_j^c)^* \\ \bar{V}_j^c(\bar{V}_j^a)^* & \bar{V}_j^c(\bar{V}_j^b)^* & |\bar{V}_j^c|^2 \end{bmatrix}_{3\times3} \tag{3.3}$$

Note that the diagonal elements are **squared** voltage magnitudes. The off-diagonal elements remain complex cross-phase voltage products.

Gan and Low define the voltage matrix as:

$$\bar{\mathbf{W}}_j = \bar{\mathbf{V}}_j\bar{\mathbf{V}}_j^H \tag{3.4}$$

Similarly:

$$\bar{\mathbf{W}}_i = \bar{\mathbf{V}}_i\bar{\mathbf{V}}_i^H \tag{3.5}$$

### 3.2.2 Solving for the right hand side of Equation 3.2

We can apply the same Hermitian transpose to the right-hand side:

$$\bar{\mathbf{W}}_j = (\bar{\mathbf{V}}_i - \bar{\mathbf{Z}}_{ij}\bar{\mathbf{I}}_{ij})(\bar{\mathbf{V}}_i^H - \bar{\mathbf{I}}_{ij}^H\bar{\mathbf{Z}}_{ij}^H)$$

**Voltage Eq. Note**

$$(\overline{\mathbf{Z}}\overline{\mathbf{I}})^H = \overline{\mathbf{I}}^H \overline{\mathbf{Z}}^H$$

The multiplication order reverses because $(AB)^H = B^H A^H$. This is a property of the conjugate transpose, not merely a rule for fixing dimensions.

Expanding the right-hand side:

$$\bar{\mathbf{W}}_j = \bar{\mathbf{V}}_i \bar{\mathbf{V}}_i^H - \bar{\mathbf{V}}_i \bar{\mathbf{I}}_{ij}^H \bar{\mathbf{Z}}_{ij}^H - \bar{\mathbf{Z}}_{ij} \bar{\mathbf{I}}_{ij} \bar{\mathbf{V}}_i^H + \bar{\mathbf{Z}}_{ij} \bar{\mathbf{I}}_{ij} \bar{\mathbf{I}}_{ij}^H \bar{\mathbf{Z}}_{ij}^H \tag{3.6}$$

We previously defined the voltage matrix in Equation 3.5 and the branch-power matrix in Equation 3.1. Accordingly, we can define the current outer-product matrix as:

$$\bar{\mathbf{L}}_{ij} = \bar{\mathbf{I}}_{ij} \bar{\mathbf{I}}_{ij}^H$$

Therefore, Equation 3.6 can be written as:

$$\boxed{\bar{\mathbf{W}}_j = \bar{\mathbf{W}}_i - \bar{\mathbf{S}}_{ij} \bar{\mathbf{Z}}_{ij}^H - \bar{\mathbf{Z}}_{ij} \bar{\mathbf{S}}_{ij}^H + \bar{\mathbf{Z}}_{ij} \bar{\mathbf{L}}_{ij} \bar{\mathbf{Z}}_{ij}^H} \tag{3.7}$$

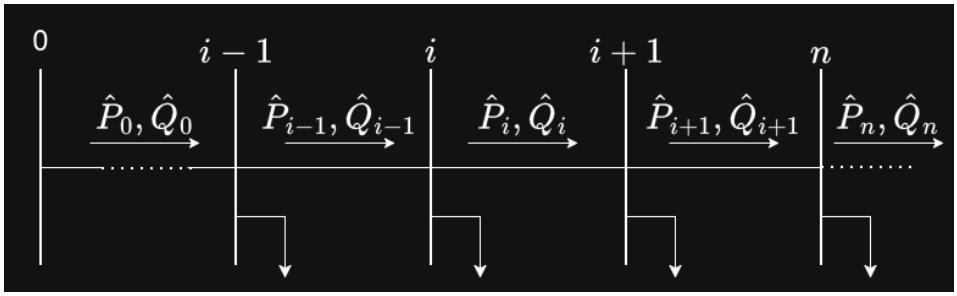
**Note**

Equation 3.7 is an exact squared-voltage **matrix** relationship derived from KVL. The term $\bar{\mathbf{Z}}_{ij} \bar{\mathbf{L}}_{ij} \bar{\mathbf{Z}}_{ij}^H$ is quadratic in line current; it is the outer product of the line voltage-drop vector with itself. Its diagonal is not the same as the real-power loss. For our immediate voltage–reactive-power work, the per-phase power equations and KVL are sufficient as starting equations, so we can keep this fuller derivation for background.

# 4 Equations Derivation

**NOTE: Baran and Wu use the DistFlow equations for a balanced, single-phase equivalent radial distribution system. We use these equations as a starting point for our unbalanced three-phase work.**

Consider the following single line diagram:




(a) ht

Figure 4.1: A single Line Diagram (SLD) for an $n$ bus system. The loads are unknown for now. The complex power ($\bar{S}$) flowing between each bus is shown.

**For each load, we consider** $P_{L,n}$ and $Q_{L,n}$, indicating real and reactive power consumed at bus $n$. Positive load means consumption.

Each line has an impedance, $\bar{Z}_n = r_n + jx_n$. The sending-end complex power flow on each line is given by:

$$\bar{S}_n = P_n + jQ_n$$

For instance, for the line leaving bus $i+1$:

$$\bar{S}_{i+1} = P_{i+1} + jQ_{i+1}$$

## 4.1 Deriving the Real and Reactive Power Equations

I think it is easier to see this if we look at the power flow between two consecutive buses. Consider the power flow between buses $n$ and $n+1$. We define $P_n$ as the power leaving bus $n$ and entering line $(n, n+1)$:

$$\underbrace{P_n}_{\text{Power leaving bus } n} = \underbrace{P_{n,loss}}_{\text{Line loss}} + \underbrace{P_{n+1}}_{\text{Power leaving bus } n+1} + \underbrace{P_{L,n+1}}_{\text{Load at bus } n+1} \tag{4.1}$$

### 4.1.1 Decomposing the Losses term in equation Equation 4.1

$$P_{loss} = r|I|^2$$

At any given bus $n$:

$$P_{n,loss} = r_n|I_n|^2 \tag{4.2}$$

Recall:

$$\bar{S} = \bar{V}\bar{I}^* = P + jQ$$

Wherein the magnitude of the apparent power is given by:

$$|S| = |I|\ |V| = \sqrt{P^2 + Q^2}$$

Squaring both sides:

$$|S|^2 = |I|^2\ |V|^2 = P^2 + Q^2$$

Solving for the squared current magnitude:

$$|I|^2 = \frac{P^2 + Q^2}{|V|^2}$$

For the line leaving bus $n$, the squared current magnitude is:

$$|I_n|^2 = \frac{P_n^2 + Q_n^2}{|V_n|^2} \tag{4.3}$$

substituting the current equation Equation 4.3 into Equation 4.2 we get:

$$P_{n,loss} = r_n \frac{P_n^2 + Q_n^2}{|\bar{V}_n|^2} \tag{4.4}$$

Substituting Equation 4.4 in the power balance between buses $n$ and $n+1$ in Equation 4.1, we get:

$$\underbrace{P_n}_{\text{Power leaving bus } n} = \underbrace{r_n \frac{P_n^2 + Q_n^2}{|\bar{V}_n|^2}}_{\text{Line loss}} + \underbrace{P_{n+1}}_{\text{Power leaving bus } n+1} + \underbrace{P_{L,n+1}}_{\text{Load at bus } n+1} \tag{4.5}$$

> **Note**
>
> Equation 4.5 is the real power DistFlow Equation. This is the exact branch real-power balance for the single-phase equivalent line model, assuming the bus has one downstream line and no additional injection. For a branching feeder, we also account for every outgoing branch.

Similarly, the reactive power can be calculated in a similar manner:

$$\underbrace{Q_n}_{\text{Q leaving bus } n} = \underbrace{x_n \frac{P_n^2 + Q_n^2}{|\bar{V}_n|^2}}_{\text{Reactive power absorbed in the line}} + \underbrace{Q_{n+1}}_{\text{Q leaving bus } n+1} + \underbrace{Q_{L,n+1}}_{\text{Q consumed by bus } n+1 \text{ load}} \tag{4.6}$$

## 4.2 Deriving the Voltage Equation:

Applying KVL to buses, $i$ and $i-1$ in Figure 4.1:

$$\bar{V}_{i-1} = \bar{Z}_{i-1}\ \bar{I}_{i-1} + \bar{V}_i \tag{4.7}$$

The current complex value can be solved for in a similar manner as in equation Equation 4.3:

$$\bar{V}_n\ \bar{I}_n^* = P_n + jQ_n$$

Taking the complex conjugate and solving for $\bar{I}_n$:

$$\bar{I}_n = \frac{P_n - jQ_n}{\bar{V}_n^*} \tag{4.8}$$

Substituting Equation 4.8 in Equation 4.7 & solving for $\bar{V}_i$:

$$\bar{V}_i = \bar{V}_{i-1} - \bar{Z}_{i-1}\ \bar{I}_{i-1} \tag{4.9}$$

$$\bar{V}_i = \bar{V}_{i-1} - \bar{Z}_{i-1}\ \frac{P_{i-1} - jQ_{i-1}}{\bar{V}^*_{i-1}}$$

We can then extend this to get only the magnitude of the voltage:

Recall $|a-b|^2 = (a-b)(a^*-b^*)$:

$$|\bar{V}_i|^2 = (\bar{V}_{i-1} - \bar{Z}_{i-1}\ \bar{I}_{i-1})\ (\bar{V}^*_{i-1} - \bar{Z}^*_{i-1}\ \bar{I}^*_{i-1})$$

Multiplying both terms, we end up with:

$$|\bar{V}_i|^2 = |\bar{V}_{i-1}|^2 - 2\ \Re\{\underbrace{\bar{V}^*_{i-1}\ \bar{I}_{i-1}}_{\substack{\text{Apparent}\\ \text{Power}\\ \text{Conjugate}}}\ \bar{Z}_{i-1}\} + |\bar{Z}_{i-1}|^2\ |\bar{I}_{i-1}|^2 \tag{4.10}$$

Recall $\bar{S}^*_{i-1} = \bar{V}^*_{i-1}\ \bar{I}_{i-1} = P_{i-1} - jQ_{i-1}$

So the middle term can be decomposed such that:

$$\bar{V}^*_{i-1}\ \bar{I}_{i-1}\ \bar{Z}_{i-1} = (P_{i-1} - jQ_{i-1})(r_{i-1} + jx_{i-1})$$

Multiplying and obtaining only the real part because we are trying to calculate the magnitude:

$$\Re\{\bar{Z}_{i-1}\ \bar{V}^*_{i-1}\ \bar{I}_{i-1}\} = r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1} \tag{4.11}$$

The last term in Equation 4.10 can be simplified further. Recall that $|\bar{Z}_{i-1}|^2 = r^2_{i-1} + x^2_{i-1}$ and $|\bar{I}_{i-1}|^2 = \frac{P^2_{i-1}+Q^2_{i-1}}{|\bar{V}_{i-1}|^2}$

$$|\bar{Z}_{i-1}|^2\ |\bar{I}_{i-1}|^2 = (r^2_{i-1} + x^2_{i-1})\ (\frac{P^2_{i-1} + Q^2_{i-1}}{|\bar{V}_{i-1}|^2}) \tag{4.12}$$

Substituting Equation 4.11 and Equation 4.12 in Equation 4.10:

$$|\bar{V}_i|^2 = |\bar{V}_{i-1}|^2 - 2(r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1}) + (r^2_{i-1} + x^2_{i-1})\ \frac{P^2_{i-1} + Q^2_{i-1}}{|\bar{V}_{i-1}|^2} \tag{4.13}$$

> **Note**
>
> Equation 4.13 is the Voltage Magnitude DistFlow Equation. This equation applies exactly to each branch under the paper's single-phase equivalent series-impedance model. It does not, by itself, describe mutual coupling in an unbalanced three-phase system.

## 4.3 Three-phase Unbalanced System

Using the same KVL we used before, Equation 4.9, we can set a system of equations for buses $i$ and $j$ such that $i$ is upstream of $j$:

$$\begin{bmatrix} \bar{V}_j^a \\ \bar{V}_j^b \\ \bar{V}_j^c \end{bmatrix} = \begin{bmatrix} \bar{V}_i^a \\ \bar{V}_i^b \\ \bar{V}_i^c \end{bmatrix} - \begin{bmatrix} \bar{Z}_{ij}^{aa} & \bar{Z}_{ij}^{ab} & \bar{Z}_{ij}^{ac} \\ \bar{Z}_{ij}^{ba} & \bar{Z}_{ij}^{bb} & \bar{Z}_{ij}^{bc} \\ \bar{Z}_{ij}^{ca} & \bar{Z}_{ij}^{cb} & \bar{Z}_{ij}^{cc} \end{bmatrix} \begin{bmatrix} \bar{I}_{ij}^a \\ \bar{I}_{ij}^b \\ \bar{I}_{ij}^c \end{bmatrix} \tag{4.14}$$

Note that the diagonal elements of $\bar{Z}_{ij}$ are self-impedances, and the off-diagonal elements are mutual impedances. This is a three-phase line model; a line with fewer phases uses the appropriate reduced impedance matrix. Neutral and transformer modeling must be handled consistently in the feeder model.

For any given phase, the voltage equation can be extracted from the above matrix form. For instance:

$$\bar{V}_j^a = \bar{V}_i^a - \bar{Z}_{ij}^{aa}\bar{I}_{ij}^a - \bar{Z}_{ij}^{ab}\bar{I}_{ij}^b - \bar{Z}_{ij}^{ac}\bar{I}_{ij}^c \tag{4.15}$$

We can rewrite the current terms in Equation 4.15 using complex power for each phase. At the sending end of line $(i, j)$:

$$\begin{aligned} \bar{S}_{ij}^a &= \bar{V}_i^a(\bar{I}_{ij}^a)^* = P_{ij}^a + jQ_{ij}^a \\ \bar{S}_{ij}^b &= \bar{V}_i^b(\bar{I}_{ij}^b)^* = P_{ij}^b + jQ_{ij}^b \\ \bar{S}_{ij}^c &= \bar{V}_i^c(\bar{I}_{ij}^c)^* = P_{ij}^c + jQ_{ij}^c \end{aligned}$$

As a vector format, we can express the three complex power equations as:

$$\begin{bmatrix} \bar{S}_{ij}^a \\ \bar{S}_{ij}^b \\ \bar{S}_{ij}^c \end{bmatrix} = \begin{bmatrix} \bar{V}_i^a(\bar{I}_{ij}^a)^* \\ \bar{V}_i^b(\bar{I}_{ij}^b)^* \\ \bar{V}_i^c(\bar{I}_{ij}^c)^* \end{bmatrix} = \begin{bmatrix} P_{ij}^a + jQ_{ij}^a \\ P_{ij}^b + jQ_{ij}^b \\ P_{ij}^c + jQ_{ij}^c \end{bmatrix} \tag{4.16}$$

> **Note**
>
> Equation 4.16 is a $3 \times 1$ vector of **per-phase complex powers**. It is not equal to a $3 \times 3$ matrix. The full branch-power matrix used by Gan and Low, including cross-phase products, is discussed in the background file. Mutual coupling is already present in Equation 4.14 through the off-diagonal impedances.

### 4.3.1 Derive the current equations

Consider the complex power for phase $a$ between two buses:

$$\bar{S}_{ij}^{a} = \bar{V}_{i}^{a} (\bar{I}_{ij}^{a})^{*}$$

Take the complex conjugate of both sides:

$$\left(\bar{S}_{ij}^{a}\right)^{*} = (\bar{V}_{i}^{a})^{*} \bar{I}_{ij}^{a}$$

Solving for $\bar{I}_{ij}^{a}$:

$$\bar{I}_{ij}^{a} = \frac{\left(\bar{S}_{ij}^{a}\right)^{*}}{(\bar{V}_{i}^{a})^{*}}$$

Which can be rewritten as:

$$\bar{I}_{ij}^{a} = \frac{P_{ij}^{a} - jQ_{ij}^{a}}{(\bar{V}_{i}^{a})^{*}} \tag{4.17}$$

Solving for phases $b$ and $c$ using the same approach:

$$\bar{I}_{ij}^{b} = \frac{P_{ij}^{b} - jQ_{ij}^{b}}{(\bar{V}_{i}^{b})^{*}} \tag{4.18}$$

$$\bar{I}_{ij}^{c} = \frac{P_{ij}^{c} - jQ_{ij}^{c}}{(\bar{V}_{i}^{c})^{*}} \tag{4.19}$$

Equation 4.17, Equation 4.18, and Equation 4.19 can be used in our KVL equation Equation 4.14, such that:

$$\begin{bmatrix} \bar{V}_j^a \\ \bar{V}_j^b \\ \bar{V}_j^c \end{bmatrix} = \begin{bmatrix} \bar{V}_i^a \\ \bar{V}_i^b \\ \bar{V}_i^c \end{bmatrix} - \bar{Z}_{ij} \begin{bmatrix} \dfrac{P_{ij}^a - jQ_{ij}^a}{(\bar{V}_i^a)^*} \\ \dfrac{P_{ij}^b - jQ_{ij}^b}{(\bar{V}_i^b)^*} \\ \dfrac{P_{ij}^c - jQ_{ij}^c}{(\bar{V}_i^c)^*} \end{bmatrix} \tag{4.20}$$

Equation 4.20 expresses the receiving-end voltage phasors as a function of the sending-end voltage phasors, branch impedance, and phase real and reactive power flows (Gan and Low 2014):

$$\boxed{\bar{\mathbf{V}}_j = \mathbf{f}(\bar{\mathbf{V}}_i, \bar{\mathbf{Z}}_{ij}, \mathbf{P}_{ij}, \mathbf{Q}_{ij})}$$

> **Note**
>
> This is an **exact branch relationship**, not yet an explicit voltage-magnitude sensitivity to inverter reactive power injection. To connect an inverter at bus $k$ to voltages throughout the feeder, we must also use network current/power balance and the inverter injection convention. Gan and Low's squared-voltage matrix derivation is in the background file for reference.

# 5 Goal

We are trying to answer the following questions in this experiment: - Can an inverter at say, house 15, raise its voltage from (e.g., 0.92) up toward a target value (1.0 p.u.)? - If house 15 injects Q, how much do the voltages at the rest of the houses change? - How much reactive power is necessary to achieve a particular household voltage target? Is this amount within the inverter capability?

## 5.1 Setup

The GridLAB-D file is set to 15 houses, with a single substation transformer and distribution transformer. There are two nodes: upstream node and node 4. The upstream node is a swing bus node. Node 4 is the main node wherein all the loads connect to.

### 5.1.1 Initial Case

None of the houses have inverters in them. The voltagge behavior is depicted at each house, as well as the source node.

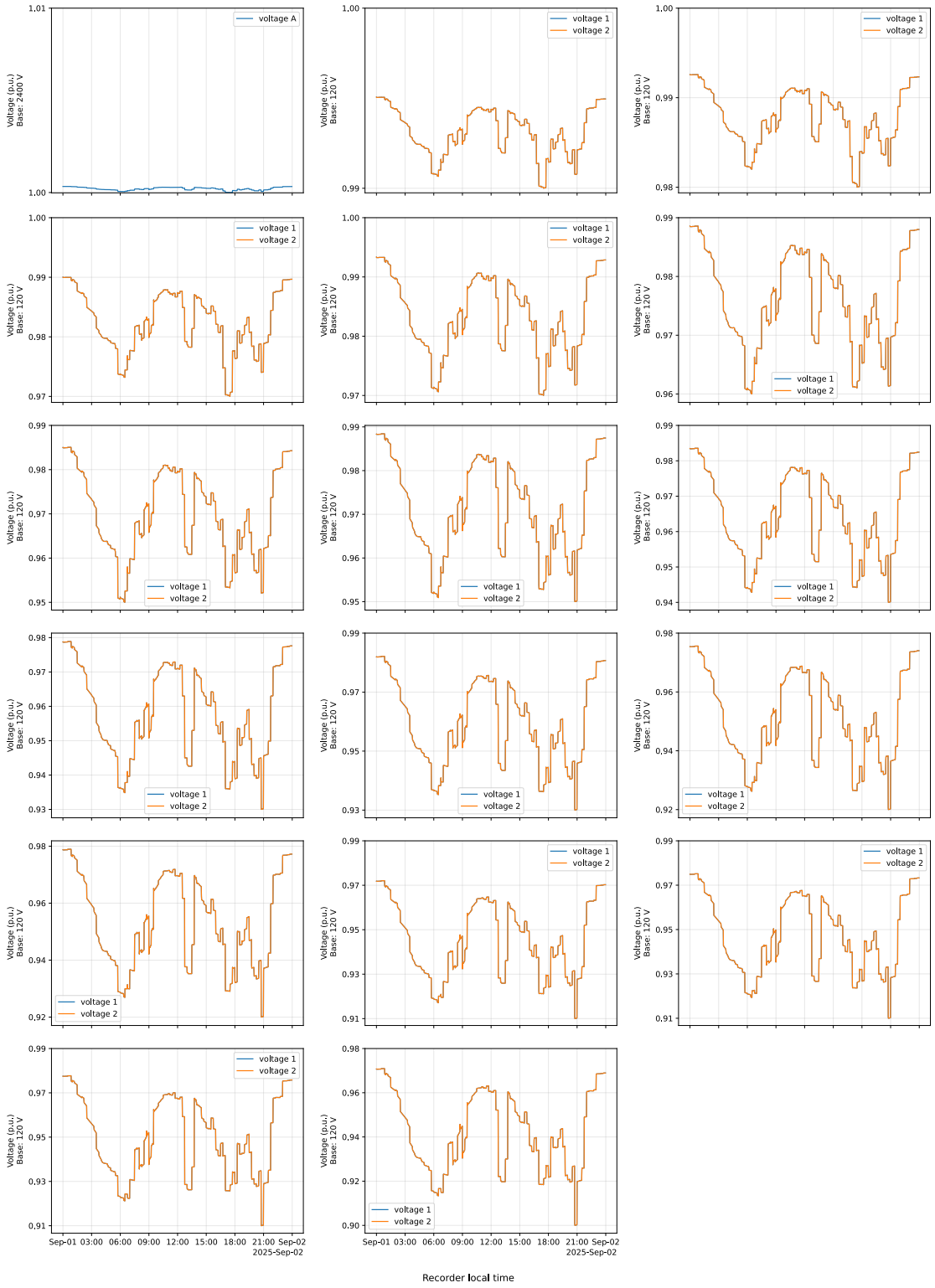



Figure 5.1: The voltage at each house, with a total of 15 houses. The first figure shows the voltage at Node 4, which is not a Swing node (obviously!)

Note that the voltage drops as the house is at a further distance from node 4.

### 5.1.2 Sweep the voltage at Node 4

When the voltage is swept at Node 4, the same voltage drop is depicted at different houses location.

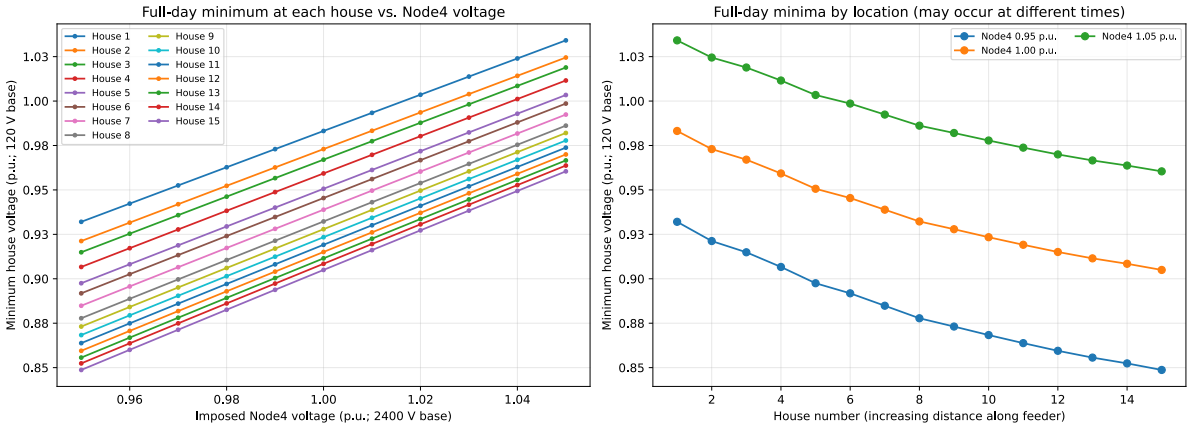



Figure 5.2: Voltage Sweeping Across Node4

Table 5.1: Simulation Results Data

| node4_pu | house | min_pu | max_pu | min_timestamp | samples |
|---|---|---|---|---|---|
| 0.95 | 10 | 0.868333 | 0.927508 | 2025-09-01 20:46:00 PDT | 1439 |
| 0.95 | 11 | 0.863792 | 0.926525 | 2025-09-01 20:46:00 PDT | 1439 |
| 0.95 | 12 | 0.859475 | 0.92555 | 2025-09-01 20:46:00 PDT | 1439 |
| 0.95 | 13 | 0.85565 | 0.925008 | 2025-09-01 20:46:00 PDT | 1439 |
| 0.95 | 14 | 0.852392 | 0.924492 | 2025-09-01 20:46:00 PDT | 1439 |
| 0.95 | 15 | 0.8487 | 0.924225 | 2025-09-01 20:46:00 PDT | 1439 |
| 0.95 | 1 | 0.932058 | 0.94515 | 2025-09-01 17:30:00 PDT | 1439 |
| 0.95 | 2 | 0.921233 | 0.942225 | 2025-09-01 17:28:00 PDT | 1439 |
| 0.95 | 3 | 0.914917 | 0.939467 | 2025-09-01 17:28:00 PDT | 1439 |
| 0.95 | 4 | 0.906658 | 0.936867 | 2025-09-01 06:29:00 PDT | 1439 |

Figure 5.2 demonstrates that the household voltage response depends on both the upstream voltage and the household's location along the feeder. Note that the distance of voltage lines in Figure 5.2 decreases consecutively. The current flow in the line supplying house 1 carries the combined current of many downstream loads. The line between house 1 and house 2 carries slightly less current, and so forth. From ohm's law:

$$\Delta\bar{\mathbf{V}} = \bar{\mathbf{Z}}\bar{\mathbf{I}}$$

If the line segments approximately equal impedances, the voltage drop across each segment generally decreases as we move downstream because the current decreases. Therefore:

$$|\Delta\bar{\mathbf{V}}_{1,2}| > |\Delta\bar{\mathbf{V}}_{13,14}| > |\Delta\bar{\mathbf{V}}_{14,15}|$$

To explain this further, let's look at all households at their lowest point, which is at 8:46 p.m.

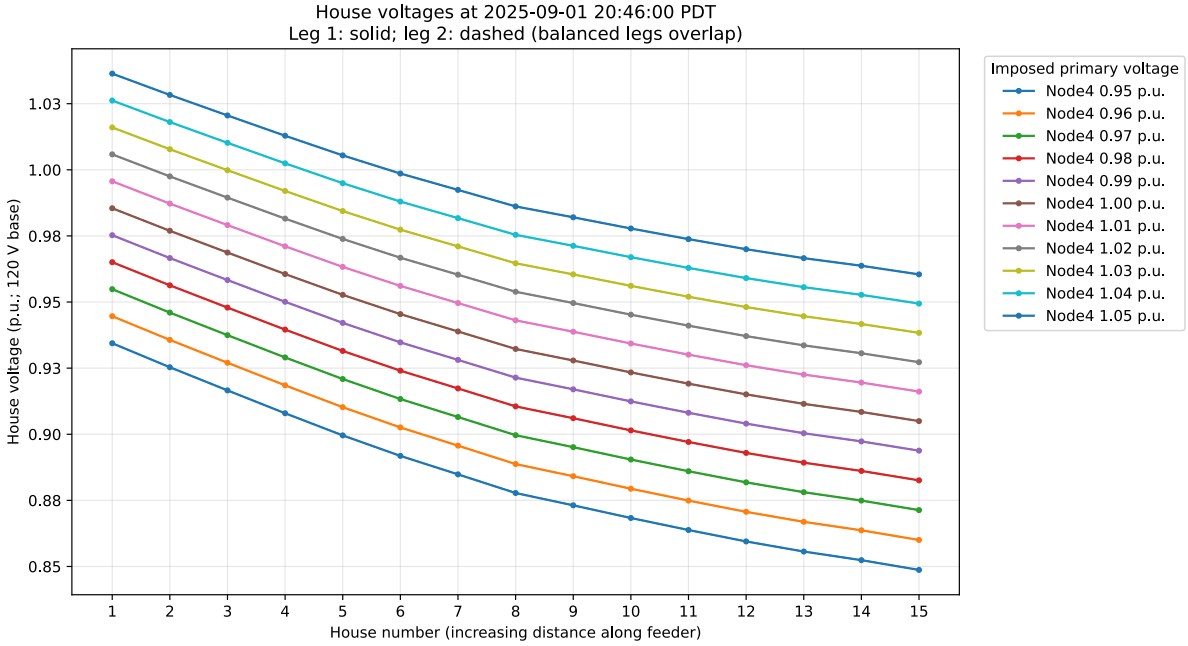



Figure 5.3: Voltage Sweep at 8:46 p.m. of all houses

The voltage behavior, as shown in Figure 5.3, is clearly nonlinear. And, the voltage drop distance between households decreases as we go down the node. This explains that the voltage trend is associated with the line impedance as well as the feeder loading, rather than a particular source voltage. This behavior is explain by:

$$\Delta\bar{\mathbf{V}} \approx \frac{RP + XQ}{\bar{\mathbf{V}}}$$

### 5.1.3 Answering the first question

The question that we try to answer in this section is:

- Can an inverter at say, house 15, raise its voltage from (e.g., 0.92) up toward a target value (1.0 p.u.)?

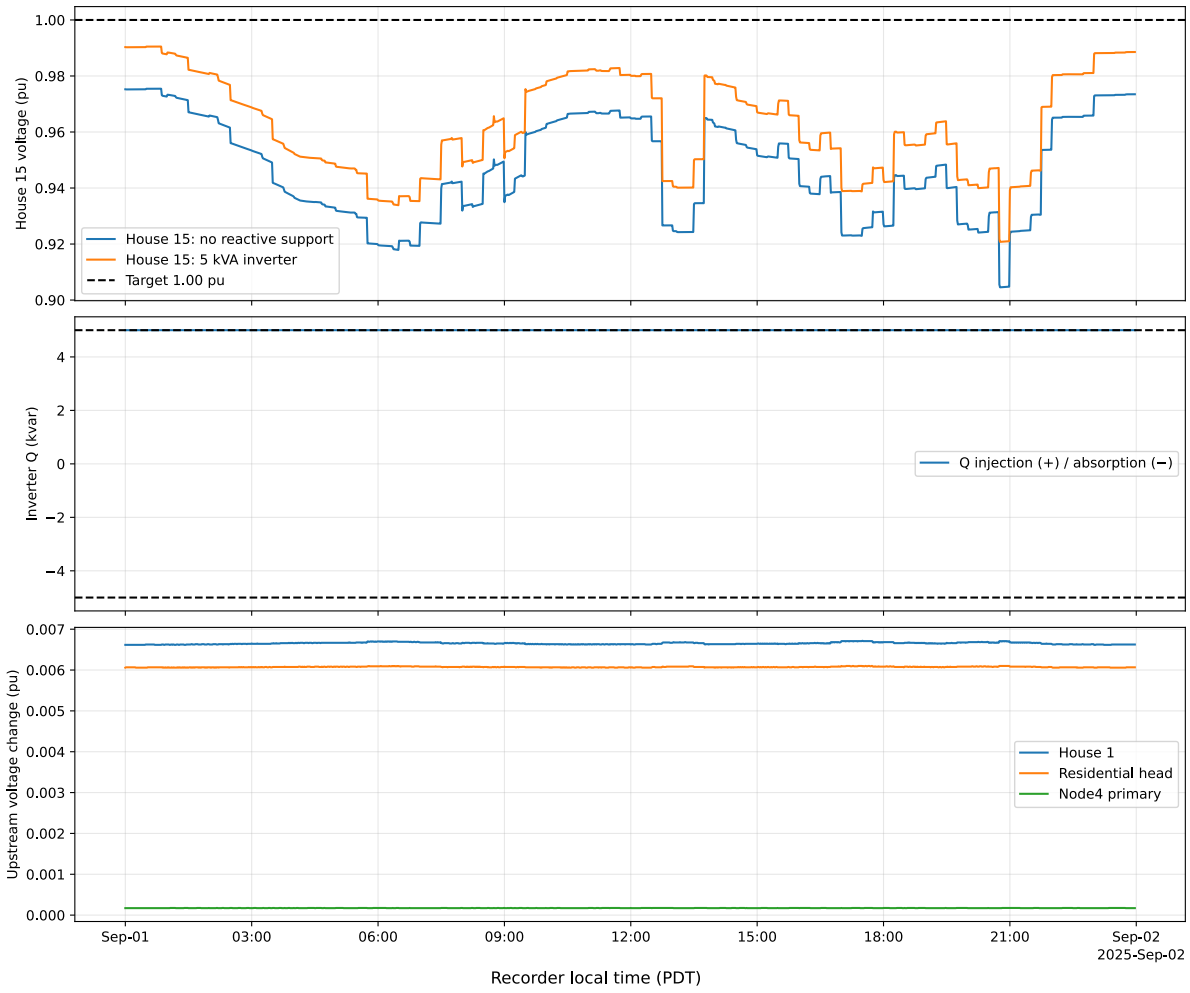



Figure 5.4: Inverter Operation at House 15

# 6 Introduction

# References

Gan, Lingwen, and Steven H Low. 2014. "Convex Relaxations and Linear Approximation for Optimal Power Flow in Multiphase Radial Networks." *2014 Power Systems Computation Conference*, 1–9.